
When Does a Shallow Topological Phase Circuit Stop Being Accurate?

Whole-Method Noise Thresholds for Adaptive Variational Berry Phases

Anonymous authors

Abstract

Adaptive variational quantum dynamics can extract an interacting many-body Berry phase with circuits far shallower than direct adiabatic Trotterization, but its demonstrated robustness has so far concerned nonadiabaticity under exact arithmetic. We determine what survives realistic execution noise. For an eight-qubit, four-site Su–Schrieffer–Heeger–Hubbard chain, we independently grow AVQITE preparations and AVQDS time-reversed loops, then freeze those trajectories and execute every circuit needed for the overlap, energy, and geometric-connection terms in a native-gate density-matrix simulator. A single noise scale jointly controls two-qubit Pauli error, readout confusion, and T_1/T_2 relaxation; every observable is independently sampled. At 8192 shots per circuit and a predeclared 0.2-rad accuracy target, the conservative mean-response crossings are $\eta = 0.02633$ for $U = 0$ and $\eta = 0.00058$ for $U = 10$, equivalent to two-qubit Pauli probabilities of 0.01316% and 0.00029%, respectively. The deeper interacting circuits therefore lose accuracy earlier. Component and single-channel experiments show that geometric quantization does not cancel independent measurement fluctuations: deterministic phase-leg biases can oppose each other, while their variances add and visibility loss makes the Hadamard phase progressively ill conditioned. The result is a device-level operating envelope, not a claim of topological protection against hardware noise.

1 Introduction

Berry phase turns a closed path of quantum states into a measurable geometric quantity with consequences ranging from polarization to topological band classification [1, 8]. This makes its direct evaluation on a quantum processor attractive: the processor represents the many-body state, follows a cyclic Hamiltonian, and interferometrically reads the accumulated phase. The obstacle is that straightforward adiabatic simulation is too deep for present hardware.

Mootz and Yao recently combined adaptive variational quantum imaginary-time evolution (AVQITE), adaptive variational quantum dynamics simulation (AVQDS), and a time-reversed cyclic protocol to obtain the Berry phase with compact circuits [6]. The method is particularly interesting because a short, visibly nonadiabatic path can still return the quantized answer. Its overlap phase φ_{qc} and a classically accumulated variational correction φ_G co-evolve, so finite-time errors cancel in their sum. That is a statement about a shared noiseless trajectory. It does not imply that errors from independently executed energy, connection, and interferometry circuits cancel. More generally, geometric and dynamical phases need not have different noise sensitivities [3].

We supply a whole-method noise characterization. “Whole method” is important here: holding φ_G exact while corrupting only the final ancilla would omit most of the measurement workload and

can yield a misleadingly favorable threshold. Instead, at every retained point on a frozen noiseless trajectory we execute noisy circuits for all Hamiltonian Paulis and all active geometric connections, in addition to both quadratures of the endpoint overlap. We then draw finite samples from each circuit. The study answers four questions:

1. At what joint device-noise scale does a 0.2-rad Berry-phase criterion first fail, with seed uncertainty?
2. How does that scale change between shallow $U = 0$ and deeper $U = 10$ circuits and between trivial and topological dimerizations?
3. How many shots remain necessary at low incoherent noise?
4. Which phase leg and physical channel causes the failure?

All thresholds, including unfavorable cases, use one preregistered rule. The code, individual seed rows, exact probabilities, construction histories, and analysis products accompany the paper.

2 Adaptive Berry-phase computation

2.1 SSHH loop and independent exact reference

We study four sites at half filling and periodic boundary conditions. With $t = 1$, staggered hopping $t_j = 1 + (-1)^j \delta$, and a spin-up boundary twist, the Hamiltonian is

$$H(\lambda) = - \sum_{j,\sigma} t_j \left(e^{i\lambda \chi_{j\sigma}} c_{j\sigma}^\dagger c_{j+1,\sigma} + \text{h.c.} \right) + U \sum_j \left(n_{j\uparrow} - \frac{1}{2} \right) \left(n_{j\downarrow} - \frac{1}{2} \right), \quad (1)$$

where $\chi_{j\sigma}$ is one only for the spin-up boundary link. Our own Jordan–Wigner construction uses interleaved spin orbitals, producing eight qubits and 18 (22) nonidentity Hamiltonian strings for $U = 0$ ($U = 10$). We use $\delta = +0.5$ (trivial), $\delta = -0.5$ (topological), and loop clock $T = 20$. At runtime, independent exact diagonalization in the half-filled sector and a 161-link Wilson loop furnish the references; no target phase or eigenstate enters variational construction.

The printed block-spin Fock seed lies in a symmetry sector orthogonal to the $U = 0, \delta > 0$ ground state. We therefore translate the same product pattern by one site for positive dimerization (for both interactions), while retaining the printed pattern for negative dimerization. This gauge/seed intervention was determined by a pre-noise sector check and is disclosed because normalized imaginary time cannot create overlap with a missing sector.

2.2 AVQITE, TETRIS growth, and signed AVQDS

The variational state is an ordered Pauli-rotation circuit

$$|\psi(\boldsymbol{\theta})\rangle = \prod_{\mu}^{\leftarrow} e^{-i\theta_{\mu} A_{\mu}} |\psi_0\rangle. \quad (2)$$

AVQITE selects from the independently enumerated 616-word UCCSD-like pool; AVQDS selects from the Hamiltonian pool. At a growth event, candidate zero-angle operators are ranked by their reduction of the normalized-state McLachlan distance and a maximal disjoint-support batch is appended. Thus each logged batch is a genuine TETRIS layer, not a depth estimate. Growth uses only the quantum-accessible metric, force, energy, and variance—never exact fidelity.

For tangent columns $|\partial_{\mu}\psi\rangle$, define

$$M_{\mu\nu} = 2 \operatorname{Re}[\langle \partial_{\mu}\psi | \partial_{\nu}\psi \rangle - \langle \partial_{\mu}\psi | \psi \rangle \langle \psi | \partial_{\nu}\psi \rangle], \quad (3)$$

$$V_{\mu}^{\text{R}} = 2s \operatorname{Im}[\langle \partial_{\mu}\psi | H | \psi \rangle - \langle \partial_{\mu}\psi | \psi \rangle E], \quad (4)$$

$$V_{\mu}^{\text{I}} = -2 \operatorname{Re}[\langle \partial_{\mu}\psi | H | \psi \rangle - \langle \partial_{\mu}\psi | \psi \rangle E]. \quad (5)$$

We solve $M\dot{\boldsymbol{\theta}} = V$ with a symmetric eigenvalue-truncated pseudoinverse. The retained relative eigenvalue cutoffs are 10^{-6} for $U = 0$ and 10^{-8} for $U = 10$; the latter avoids false pool exhaustion

in the larger interacting manifold. AVQITE and AVQDS both use $L_{\text{cut}}^2 = 10^{-4}$. RK4 steps are capped by $\max_{\mu} |\Delta\theta_{\mu}| \leq 0.01$ and $ds \leq 0.02$.

The signed clock first propagates $\lambda : 0 \rightarrow \pi$ with $s = +1$ and then $\lambda : \pi \rightarrow 2\pi$ with $s = -1$, cancelling the ideal dynamical phase. The final estimate is

$$\widehat{\varphi}_B = \widehat{\varphi}_{\text{qc}} + \widehat{\varphi}_{G,1} + \widehat{\varphi}_{G,2}, \quad (6)$$

$$\varphi_{G,1} = - \int_0^T s(t) \langle H(t) \rangle dt, \quad (7)$$

$$\varphi_{G,2} = - \int_C \sum_{\mu} a_{\mu} d\theta_{\mu}, \quad a_{\mu} = \text{Im} \langle \psi | \partial_{\mu} \psi \rangle. \quad (8)$$

For our $e^{-i\theta A}$ convention, $a_{\mu} = -\langle A_{\mu} \rangle$ on the prefix ending at gate μ . Equation (8) fixes a sign ambiguity that otherwise doubles, rather than removes, the phase of an exact e^{-iHt} test. We retain an unwrapped decomposition and use only the final principal circular phase for errors.

The endpoint overlap $z = \langle \psi(0) | \psi(T) \rangle$ is measured with two independent ancilla quadratures and

$$\widehat{\varphi}_{\text{qc}} = \text{atan2}(\widehat{\text{Im}} z, \widehat{\text{Re}} z). \quad (9)$$

Unlike a cosine-only inverse, this estimator is locally well conditioned at both quantized targets unless noise collapses $|z|$.

2.3 Frozen device quadrature

The noiseless RK4 trajectory, ansatz, and $\dot{\theta}$ are frozen before noise; we do not reconstruct noisy M or V . We record every accepted endpoint as a fine classical reference. Eighty-one uniform points, the two sides of every growth event, and both sides of the time-reversal cusp are mandatory device nodes. A deterministic greedy refinement adds endpoints until the signed integral error in each φ_G leg is below 0.005 rad relative to the full endpoint path. The retained grid must also agree with the independently RK4- integrated total phase within 0.02 rad.

Energy uses nonuniform time-trapezoid weights. For the connection leg we use the exactly equivalent discrete frozen-velocity form

$$\widehat{\varphi}_{G,2} = -\frac{1}{2} \sum_i [\widehat{\mathbf{a}}_i + \widehat{\mathbf{a}}_{i+1}]^T (\boldsymbol{\theta}_{i+1} - \boldsymbol{\theta}_i). \quad (10)$$

Thus device noise enters every connection estimate while the secant $\Delta\theta/\Delta t$ remains noiseless and frozen. This path form avoids amplifying coordinate-velocity spikes of the gauge-singular metric. The grid rule was fixed after disclosed channel-free convergence and runtime checks and before any nonzero-noise result.

3 Noise model and statistical protocol

Each Pauli rotation is compiled into parallel single-qubit basis-change layers, a serialized all-to-all CNOT parity ladder, $R_z(2\theta)$, and the inverse ladder. A strength η scales a concrete reference profile:

$$p_2 = 0.005\eta, \quad p_m = 0.02\eta, \quad T_1 = 100/\eta \mu\text{s}, \quad T_2 = 80/\eta \mu\text{s}. \quad (11)$$

Here p_2 is the total probability of a uniformly selected nonidentity two-qubit Pauli after every CNOT (average gate infidelity $0.8p_2$), and p_m is independent symmetric bit confusion on every measured qubit. Native single-qubit and CNOT durations are 35 ns and 300 ns. After every scheduled layer, zero-temperature amplitude damping and pure dephasing consistent with T_1, T_2 act on every qubit, including idles. The density matrix is propagated through the actual eight-qubit observable circuits and nine-qubit controlled-overlap circuits; noise is never added to the final phase as a surrogate.

At every retained node we independently prepare the noisy prefix needed for each a_{μ} , prepare the full noisy state for each Hamiltonian Pauli, append its noisy basis rotation and readout, and draw a binomial sample. The two Hadamard quadratures are independent. “8192 shots” therefore means 8192 shots for *each* such circuit. Exact CPTP probabilities are cached only to avoid repeating

deterministic density propagation across random seeds; each of 24 seeds, 4100–4123, draws fresh counts for every observable.

The complete original joint sweep is

$$\eta \in \{0, .025, .05, .075, .1, .15, .2, .3, .4, .6, .8, 1\}. \quad (12)$$

After it bracketed the interacting crossings inside its first interval, we retained every original point and uniformly added $\eta \in \{0.00025, 0.0005, 0.00075, 0.001, 0.00125\}$ for all four configurations. This disclosed resolution amendment changed neither seeds, shots, tolerance, nor the extraction rule. We measure circular error $d(\hat{\varphi}, \varphi^{\text{ED}}) = |\arg e^{i(\hat{\varphi} - \varphi^{\text{ED}})}|$. A point passes when its mean error plus 1.96 standard errors is at most 0.2 rad. The discrete threshold is the last consecutively passing grid point before the first failure; re-entry is forbidden. For a continuous estimate, weighted nondecreasing isotonic regression of mean error is linearly interpolated at 0.2 rad. We report a 95% interval from 2000 paired-seed bootstrap resamples. The threshold for a given U is the smaller of its two dimerization thresholds.

We evaluate phase discrimination on the same circle rather than subtracting principal values across the branch cut. For equal-seed trivial/nontrivial pairs, let $d_c(a, b) = |\arg e^{i(a-b)}|$ and define the inter-regime separation $\Delta_{U,s} = d_c(\hat{\varphi}_{0,s}, \hat{\varphi}_{\pi,s})$. Our signed nearest-prototype margin is

$$m_{U,s} = \min_{r \in \{0, \pi\}} \frac{d_c(\hat{\varphi}_{r,s}, \varphi_r^{\text{ED}}) - d_c(\hat{\varphi}_{r,s}, \varphi_r^{\text{ED}})}{2} = \frac{\pi}{2} - \max_r d_c(\hat{\varphi}_{r,s}, \varphi_r^{\text{ED}}), \quad (13)$$

where the equality uses the antipodal ED prototypes. Thus $m_{U,s} > 0$ only when both estimates are closer to their own prototype than to the other one; zero is the circular decision boundary. We report the paired mean, its mean-minus-1.96-SE lower bound, and the minimum observed seed margin.

Finite sampling is a separate axis. At fixed $\eta = 2.5 \times 10^{-4}$, seeds 5100–5123 sample 128 through 16384 shots per circuit. The shot floor is the smallest passing count for which all larger counts pass. Finally, at $\eta \in \{0.1, 0.3, 0.6\}$ we rerun two-qubit depolarization, relaxation, and readout separately, alongside joint noise, and retain $\varphi_{\text{qc}}, \varphi_{G,1}, \varphi_{G,2}$ per seed.

4 Results

4.1 Noiseless anchors and resource scaling

Table 1: Noiseless anchors and the last consecutively passing sampled noise point. Noisy phases are circular means and errors are means of per-seed circular errors. The threshold is the continuous isotonic crossing in joint scale η with a paired-seed 95% bootstrap interval; $p_2 = 0.005\eta$. Configuration-level threshold and device-unit values are repeated in both setting rows so that every entry is explicit.

(U, δ)	setting	$\bar{\varphi}_B$	error	η_* [95% CI]	$p_{2,*}$ (%)	depth	N_{CX}
(0, +0.5)	noiseless	0.0001	0.0001	0.02633 [0.02465, 0.02773]	0.01316	12	108
(0, +0.5)	$\eta = 0.00125$	-0.0213	0.0289	0.02633 [0.02465, 0.02773]	0.01316	12	108
(0, -0.5)	noiseless	3.1415	0.0001	0.03199 [0.03045, 0.03337]	0.01599	18	188
(0, -0.5)	$\eta = 0.02500$	-2.9829	0.1587	0.03199 [0.03045, 0.03337]	0.01599	18	188
(10, +0.5)	noiseless	0.0000	0.0000	0.00106 [0.00087, 0.00118]	0.00053	68	684
(10, +0.5)	$\eta = 0.00075$	0.1590	0.1590	0.00106 [0.00087, 0.00118]	0.00053	68	684
(10, -0.5)	noiseless	-3.1397	0.0019	0.00058 [0.00048, 0.00065]	0.00029	65	646
(10, -0.5)	$\eta = 0.00025$	-3.0487	0.1119	0.00058 [0.00048, 0.00065]	0.00029	65	646

All four independently grown circuits recover the ED branch and topological label before noise (Table 1). The maximum instantaneous state infidelity can be substantial on the short interacting topological loop, yet Eq. (6) remains accurate: this reproduces the intended nonadiabatic phase cancellation rather than assuming adiabatic following. The logged selections also reproduce the central resource hierarchy: the $U = 10$ ansatz is several times deeper and more entangling than

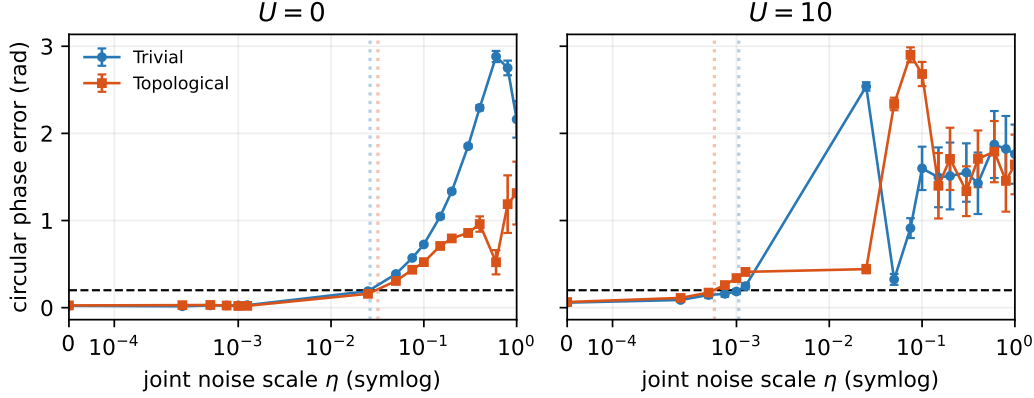


Figure 1: Whole-method circular Berry-phase error versus joint noise scale at 8192 shots per circuit. Markers are 24-seed means, bars are 95% normal intervals of the mean, horizontal dashed lines are the predeclared 0.2-rad tolerance, and dotted vertical lines are isotonic crossings. A symmetric log abscissa resolves the channel-free point and the low-noise refinement.

$U = 0$, and the negative-dimerization $U = 0$ circuit is costlier than its trivial partner. At $U = 10$, however, our two independently grown resources are nearly tied and slightly reversed (68 layers/684 CX for trivial versus 65/646 for topological). We therefore make no topology-dependent resource-scaling claim from this single adaptive realization; the robust comparison is shallow $U = 0$ versus deep $U = 10$.

As an external trace check only, the supplied figshare data at the same (δ, T) fold to 0 and π and contain 15/35 layers for trivial/topological $U = 0$ and 71/143 for $U = 10$. We did not load their Hamiltonian or stored ansatz. Differences in exact layer counts are expected from adaptive ties, seed gauge, and the stabilized pseudoinverse; our resource numbers come solely from our chronological operator log.

4.2 A depth-dependent whole-method threshold

Figure 1 is the primary resilience result. The conservative mean-response crossings are $\eta = 0.02633$ for the shallow noninteracting pair and $\eta = 0.00058$ for the interacting pair, limited by the trivial and topological configurations, respectively. In device units these correspond to total post-CNOT Pauli probabilities below approximately 0.01316% and 0.00029%. The corresponding readout-error ceilings are 0.05266% and 0.00116%, while the effective (T_1, T_2) floors are $(3798, 3039) \mu\text{s}$ and $(172882, 138305) \mu\text{s}$, respectively. These numbers are profile coordinates that co-vary, not independent single-channel limits. Table 1 reports each configuration separately, including the last passing point and bootstrap uncertainty. At the largest sampled scale strictly below each conservative mean-response crossing, $\eta = 0.02500$ for $U = 0$ and $\eta = 0.00050$ for $U = 10$, the mean paired circular separations are 2.793 and 3.054 rad, with mean-minus-1.96-SE bounds of 2.779 and 3.025 rad. More stringently, the signed margins in Eq. (13) are 1.375 and 1.369 rad, with lower bounds 1.364 and 1.342 rad and minimum observed seed margins 1.328 and 1.186 rad. All 24/24 and 24/24 paired replicas therefore remain on the correct side of the circular class boundary. The 0.2-rad accuracy rule is substantially stricter than merely assigning the nearest of 0 and π . The two reported grid notions should not be conflated: η_* estimates where the population mean reaches 0.2 rad, whereas each under-noise row is the last sampled point satisfying the stricter mean-plus-1.96-SE certification. The bootstrap interval quantifies uncertainty in the former.

The depth trend is visible but not reducible to endpoint fidelity alone (Fig. 2). Noise also changes the measured energy and connection integrands at every path node. Two states with similar endpoint infidelity can therefore have different accumulated $\varphi_{G,1}$ drift, $\varphi_{G,2}$ attenuation, and overlap visibility.

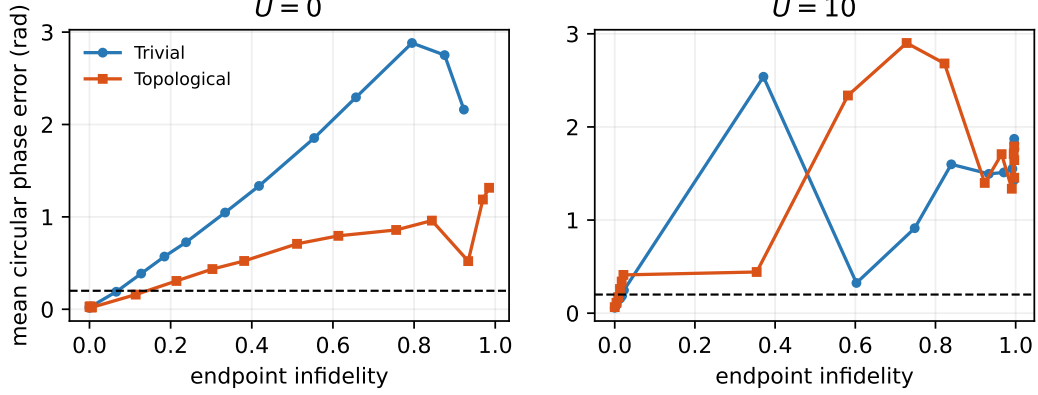


Figure 2: Mean phase error against endpoint state infidelity along the joint sweep. Infidelity is useful but not a sufficient phase-failure proxy: the estimator combines an endpoint overlap with two path integrals.

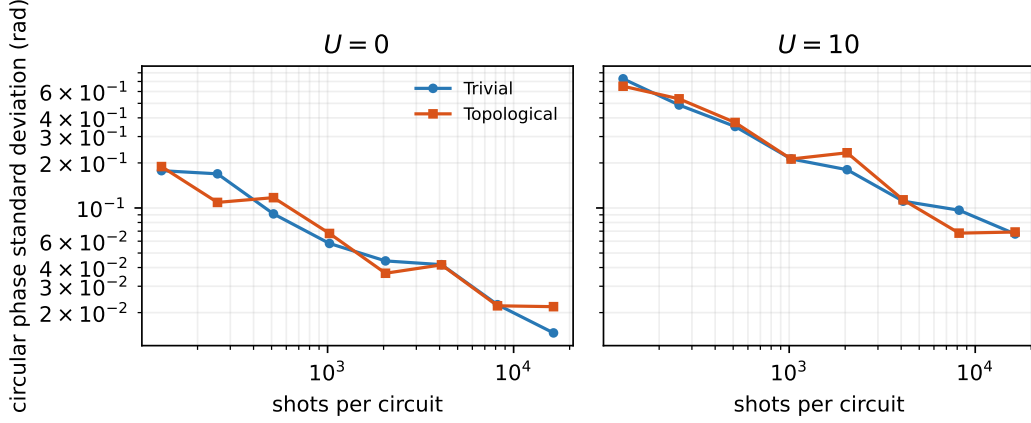


Figure 3: Circular standard deviation versus independent shots per circuit at $\eta = 2.5 \times 10^{-4}$. Every point resamples all energy, connection, and overlap circuits with 24 fresh seeds.

4.3 Sampling floor

At low incoherent noise, the minimum passing counts are 128 ($U = 0$, trivial), 256 ($U = 0$, topological), 2048 ($U = 10$, trivial), and 4096 ($U = 10$, topological) shots per circuit. The trend in Fig. 3 is compatible with binomial $S^{-1/2}$ contraction: fitted log-log slopes are -0.518 , -0.459 , -0.488 , and -0.507 in the displayed configuration order. The configuration-dependent prefactors grow with the number of independently sampled active connection terms. Increasing shots removes that stochastic spread, not the infinite-shot channel bias; this is why a shot floor and an incoherent-noise threshold are distinct specifications. The originally registered $\eta = 0.025$ shot sweep and an initial $\eta = 0.001$ pilot are retained in the logs; both are censored for at least one interacting configuration because incoherent bias has already crossed the tolerance. We therefore use 2.5×10^{-4} for the sampling floor, selected before its shot-floor seeds were drawn; no joint-sweep datum or threshold rule was changed.

5 Why the cancellation fails

Write each sampled leg as its noiseless value plus bias and fluctuation, $\hat{\varphi}_k = \varphi_k + b_k + \epsilon_k$. The shared frozen path can make $b_{qc} + b_{G,2}$ small, just as finite- T contributions cancel in the noiseless

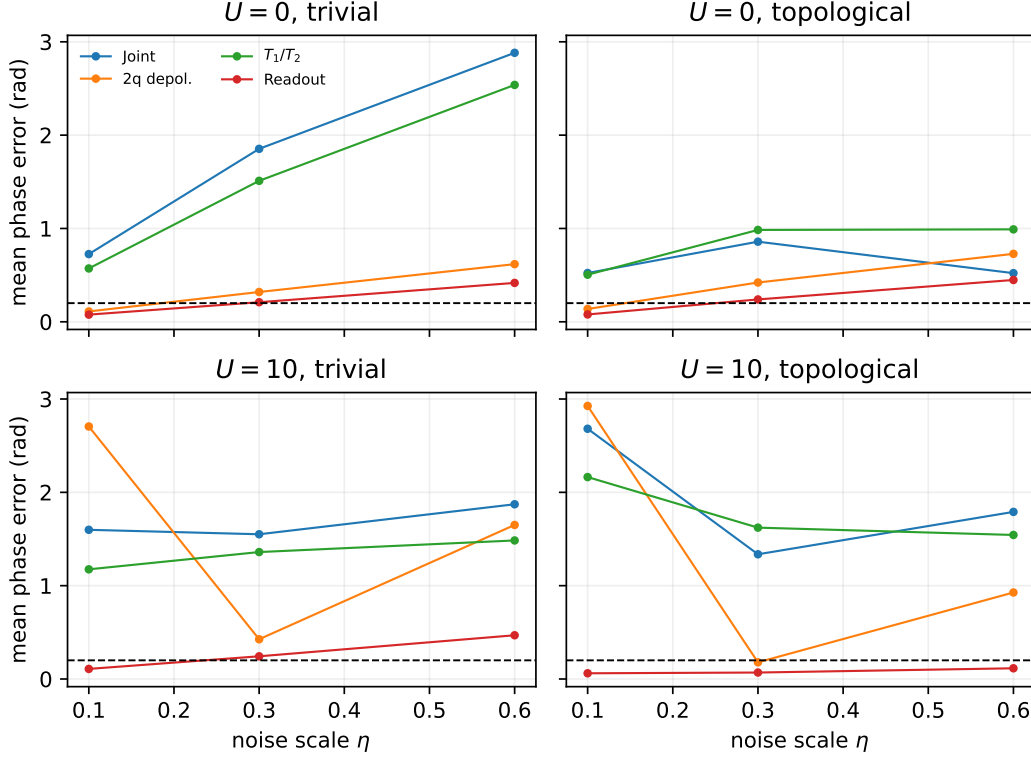


Figure 4: Channel-isolation experiments at the three preregistered strengths. Each curve still samples the complete $\varphi_{qc} + \varphi_{G,1} + \varphi_{G,2}$ estimator; only the indicated physical channel is enabled.

theory. But the three legs are obtained from disjoint circuit shots, so, conditional on the trajectory,

$$\text{Var}(\hat{\varphi}_B) = \sum_k \text{Var}(\epsilon_k) + 2 \sum_{k < \ell} \text{Cov}(\epsilon_k, \epsilon_\ell), \quad (14)$$

and the expected sampling covariances vanish. The empirical covariance term in our per-seed logs fluctuates around zero rather than providing systematic cancellation. Topological quantization constrains the ideal closed path; it does not correlate independent Bernoulli draws.

The phase legs have different failure physics. Depolarizing and readout channels contract nonidentity Pauli expectations, directly biasing both energy and connection integrals toward their identity-only limits. Relaxation is not unital: it drives the occupation-encoded state toward computational zero while its accumulation follows scheduled native depth. The energy leg is especially distinct: $\varphi_{G,1}$ integrates this distorted expectation over the full loop, so imperfect forward/return cancellation becomes a coherent drift rather than a final-readout fluctuation. Meanwhile the controlled endpoint circuit loses visibility; since the uncertainty of an angular estimate scales inversely with radius in the $(\text{Re } z, \text{Im } z)$ plane, its atan2 phase becomes noisy as $|z| \rightarrow 0$.

Figure 4 and the logged three-leg decomposition distinguish these mechanisms quantitatively. At the diagnostic strength $\eta = 0.1$, thermal-only mean errors are 0.572 and 0.505 rad for the two $U = 0$ circuits, versus 0.111/0.137 rad for depolarization and 0.077/0.079 rad for readout alone. In the deeper $U = 10$ circuits, depolarization-only errors rise to 2.706/2.926 rad and thermal-only errors to 1.175/2.164 rad, whereas readout alone remains at 0.108/0.061 rad. Joint-noise endpoint visibility has already collapsed to 0.0235/0.0221 there. Moreover, twice the summed empirical cross-covariance of the three phase legs is +0.055 and +0.004 rad^2 , respectively: finite-sample covariance provides no protective negative term. No one cancellation therefore protects all channels. An opposing mean bias can temporarily flatten the joint curve, but independently sampled variance, energy-leg drift, and visibility loss persist. This explains the finite threshold and its depth dependence without equating nonadiabatic robustness with hardware-noise robustness.

6 Failure analysis, cost, and limitations

Several numerical failures were informative. A cosine-only ancilla estimator is singular at the desired $0/\pi$ targets; using both quadratures removes that avoidable pathology. A 41-node and then an 81-node uniform connection quadrature missed sharp coordinate features even while physical-state fidelity was smooth. Truncating gauge-null metric directions and integrating the connection along the actual parameter path repaired the issue. All rejected runs, including wrong-sector preparation, overstrict variational cutoffs, and quadrature variants, remain in `proposal/attempts_log.md`; they are not silently omitted from the evidence base.

Physical failure is visibly nonmonotone after the first crossing because the reported quantity is circular. For example, the $U = 10$ trivial mean error is 2.539 rad at $\eta = 0.025$ but wraps back to 0.324 rad at $\eta = 0.05$, where the nearest-quantized-value label happens to be correct for all 24 seeds even though overlap visibility is only 0.134. This is aliasing, not restored accuracy. Our consecutive-pass rule therefore rejects every such re-entry; it also illustrates why a binary topological label alone is an unsafe resilience metric.

Whole-method sampling is expensive even for eight qubits. For each node, the inventory contains every Hamiltonian Pauli and every active connection prefix, plus two endpoint quadratures: one replica uses 3839/3990 circuits for the two $U = 0$ cases and 20310/19657 for $U = 10$. Across the main sweep our 24-seed accounting represents 159,750,291,456 physical shots. The simulator saves one exact CPTP probability tensor per configuration/noise point and performs genuine binomial draws for every seed; this changes computation time, not the statistical experiment. The four independent anchors consumed 0.28 summed CPU-hours, the joint probability sweep 10.22 CPU-hours, and channel isolation 3.84 CPU-hours; wall time was reduced by process-level parallelism. Measurement grouping, optimal shot allocation, and error mitigation [2] could reduce or redistribute this cost, but none is used in the threshold claimed here.

The scope is deliberately narrow in four ways. First, $N = 4$ and all-to-all connectivity do not predict a routing-heavy device. Second, the Markovian profile omits coherent calibration error, crosstalk, leakage, nonzero thermal population, and drift. Third, noise acts on execution of a noiselessly grown trajectory; noisy estimation of M, V could change ansatz growth and is a harder problem. Fourth, a one-parameter joint scale is a profile-dependent operating envelope, not a universal hardware constant. The channel-isolation runs expose sensitivity directions, but extrapolation requires a device’s own rates and timing. These limitations make the reported thresholds conservative evidence for this precisely specified pipeline rather than a broad claim of fault tolerance.

7 Related work

AVQDS and AVQITE provide the adaptive construction used here [10, 4]; compressed/TETRIS variants emphasize shallower circuits, measurement economy, and stable treatment of singular metrics [11]. Gate-based Berry-phase readout and robust topology protocols have been developed in complementary settings [7, 9]. Those works motivate an expectation that geometry or quantization can help, but they do not provide a whole-method hardware-noise threshold for the present co-evolving AVQDS estimator. Our study also complements a recent analysis showing that paired $\pm H$ protocols cancel leading deterministic adiabatic errors and can further suppress their residuals by extrapolation or runtime randomization [5]. Those controls target finite-runtime bias, whereas our limiting errors arise from separately executed noisy observable circuits. Our contribution is therefore not a new variational algorithm: it is an empirical operating envelope and a phase-leg explanation for an existing one.

8 Conclusion

Adaptive compression makes many-body Berry-phase estimation shallow enough that a hardware-noise threshold is worth asking for; it does not make the answer automatically robust. On the frozen SSHH trajectories studied here, the shallow and deep regimes have distinct device envelopes, and finite-shot requirements remain even at nearly ideal incoherent noise. The key diagnostic is to treat Eq. (6) as three separately executed estimators. Shared trajectory biases may cancel, independent

fluctuations do not, and the dynamical energy leg accumulates its own drift. Reporting all three legs, visibility, and state infidelity is consequently more informative than a single noisy topological label.

Reproducibility statement

The implementation is self-contained under `proposal/code`. The pre-result method and every disclosed amendment are in `proposal/logs/method_preregistration.json`. Exact references, chronological operator selections, resource traces, density-matrix probabilities, every finite-shot seed row, threshold bootstrap output, the paired circular discrimination margins, and software versions are retained under `proposal/logs`. Reference archive inspection is documented separately and archived `ansatz/model` files were not used as inputs. Unit tests cover the Jordan–Wigner model, tangent identities, McLachlan dynamics, adaptive path quadrature, native compilation, CPTP channels, relaxation timing, readout confusion, and sampling.

References

- [1] M. V. Berry. Quantal phase factors accompanying adiabatic changes. *Proceedings of the Royal Society of London A*, 392:45–57, 1984.
- [2] Zhenyu Cai, Ryan Babbush, Simon C. Benjamin, Suguru Endo, William J. Huggins, Ying Li, Jarrod R. McClean, and Thomas E. O’Brien. Quantum error mitigation. *Reviews of Modern Physics*, 95:045005, 2023.
- [3] R. K. L. Colmenar, Utkan Güngördü, and J. P. Kestner. Conditions for equivalent noise sensitivity of geometric and dynamical quantum gates. *PRX Quantum*, 3:030310, 2022.
- [4] Niladri Gomes, Anirban Mukherjee, Feng Zhang, Thomas Iadecola, Cai-Zhuang Wang, Kai-Ming Ho, Peter P. Orth, and Yong-Xin Yao. Adaptive variational quantum imaginary time evolution approach for ground state preparation. *Advanced Quantum Technologies*, 4:2100114, 2021.
- [5] Chusei Kiumi. Adiabatic error cancellation in berry phase estimation, 2026.
- [6] Martin Mootz and Yong-Xin Yao. Efficient berry phase calculation via adaptive variational quantum computing approach. *APL Quantum*, 3(1):016111, 2026.
- [7] Bruno Murta, Gonçalo Catarina, and Joaquín Fernández-Rossier. Berry phase estimation in gate-based adiabatic quantum simulation. *Physical Review A*, 101:020302, 2020.
- [8] Raffaele Resta. Macroscopic polarization in crystalline dielectrics: The geometric phase approach. *Reviews of Modern Physics*, 66:899–915, 1994.
- [9] Xiao Xiao, J. K. Freericks, and A. F. Kemper. Robust measurement of wave function topology on NISQ quantum computers. *Quantum*, 7:987, 2023.
- [10] Yong-Xin Yao, Niladri Gomes, Feng Zhang, Cai-Zhuang Wang, Kai-Ming Ho, Thomas Iadecola, and Peter P. Orth. Adaptive variational quantum dynamics simulations. *PRX Quantum*, 2:030307, 2021.
- [11] Feng Zhang, Cai-Zhuang Wang, Thomas Iadecola, Peter P. Orth, and Yong-Xin Yao. Adaptive variational quantum dynamics simulations with compressed circuits and fewer measurements. *Physical Review B*, 111:094310, 2025.